ASSIGNMENT - 4

Introduction to Asymmetric Cryptography

Cryptography and Network Security

Instructions

- Attempt all 8 questions. Marks are indicated against each question.
- Support theoretical answers with relevant examples, algorithm steps and diagrams.
- Use neat, labelled diagrams wherever required.
- Write answers in your own words. Cite any reference books or websites you use.

Questions

Q1. *[Overview and principles]* **[10 Marks]**

Explain the basic principles of asymmetric (public-key) cryptography. Describe the roles of the public key and the private key, and explain how the same scheme can provide confidentiality, authentication and non-repudiation. Illustrate your answer with a neat labelled diagram of the encryption/decryption process between two parties, Alice and Bob.

Ans- Introduction

Asymmetric cryptography, also called **public-key cryptography**, is a method of encryption that uses **two different but mathematically related keys**:

- **Public Key**
- **Private Key**

The public key can be shared with everyone, while the private key must be kept secret by its owner.

Roles of the Keys

- **Public Key:** Used by others to encrypt information for the owner or to verify the owner's digital signature.
- **Private Key:** Kept secret and used to decrypt information or create a digital signature.

Providing Confidentiality

Suppose Alice wants to send a secret message to Bob:

1. Alice obtains Bob's public key.
2. Alice encrypts the message using Bob's public key.
3. Bob receives the ciphertext.
4. Bob decrypts it using his private key.

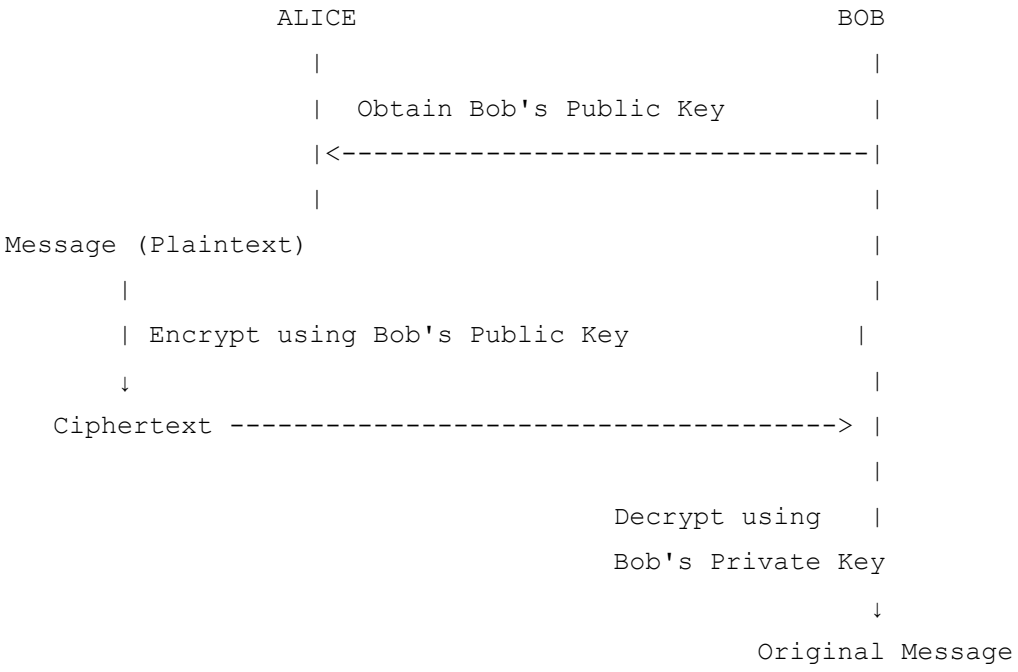
Only Bob's private key can normally decrypt the message.

Providing Authentication and Non-Repudiation

For authentication, Alice can create a **digital signature** using her private key. Bob uses Alice's public key to verify the signature.

- **Authentication:** Bob can check that the message came from Alice.
- **Non-repudiation:** A properly implemented digital signature provides evidence that the message was signed by the holder of Alice's private key.

Labelled Diagram



Q2. *[Overview and techniques]* **[8 Marks]**

Differentiate between symmetric and asymmetric cryptography under the following headings: number of keys, key distribution, speed, key length, scalability for n users, and typical applications. Present your answer in tabular form.

Ans-

Heading	Symmetric Cryptography	Asymmetric Cryptography
Number of Keys	Uses one shared secret key	Uses two keys: public and private
Key Distribution	Secret key must be securely shared	Public key can be openly distributed
Speed	Faster	Slower
Key Length	Usually shorter keys provide strong security	Usually requires much larger keys
Scalability for n Users	Difficult because many shared keys are needed	Easier because each user can have one key pair
Typical Applications	Bulk data encryption, file encryption, disk encryption	Key exchange, digital signatures, authentication
Examples	AES, ChaCha20	RSA, ECC

Main Difference

The biggest difference is the way keys are used. Symmetric cryptography uses **the same secret key** for encryption and decryption. Asymmetric cryptography uses **a public-private key pair**.

Conclusion

Symmetric encryption is generally preferred for encrypting large amounts of data because it is fast. Asymmetric cryptography is mainly useful for **secure key exchange, authentication and digital signatures**.

Q3. *[Overview and techniques]* **[10 Marks]**

Describe the main techniques and building blocks used in public-key cryptography, including trapdoor one-way functions, the Diffie-Hellman key exchange, digital signatures, and public-key infrastructure (PKI) with digital certificates. Explain briefly how each technique contributes to secure communication.

Ans- Main Techniques and Building Blocks of Public-Key Cryptography

Public-key cryptography uses several important mathematical techniques and security mechanisms.

1. Trapdoor One-Way Functions

A one-way function is easy to calculate in one direction but very difficult to reverse.

A **trapdoor** provides special information that makes reversing the function practical for an authorized user.

Importance

- Forms the mathematical foundation of many public-key systems.
- Helps create secure encryption and digital signatures.

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2. Diffie-Hellman Key Exchange

Diffie-Hellman allows two parties to establish a **shared secret over an insecure network**.

Basic Process

- Alice creates a private value.
- Bob creates a private value.
- They exchange public values.
- Each uses their private value with the other party's public value.
- Both calculate the same shared secret.

The secret itself is not directly transmitted.

Importance

It allows secure establishment of a shared key.

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3. Digital Signatures

A digital signature is created using the sender's **private key** and verified using the sender's **public key**.

It provides:

- Authentication
- Integrity
- Evidence of who signed the message

Q4. *[Historical development]* **[10 Marks]**

Trace the historical development of public-key cryptography. Your answer should cover (a) the key distribution problem that motivated it, (b) the contribution of Diffie and Hellman (1976), (c) the invention of RSA (1977) by Rivest, Shamir and Adleman, (d) the earlier classified work at GCHQ by Ellis, Cocks and Williamson, and (e) later schemes such as ElGamal and Elliptic Curve Cryptography.

Ans- 1. Key Distribution Problem

Before public-key cryptography, symmetric systems required both parties to have the same secret key. The main problem was **how to securely share that key** over an insecure network.

This key distribution problem motivated the development of public-key techniques.

2. Diffie and Hellman – 1976

In **1976**, Whitfield Diffie and Martin Hellman introduced the idea of public-key cryptography and proposed the **Diffie-Hellman key exchange**.

Their work showed that two people could establish a shared secret without directly sending the secret key.

3. RSA – 1977

In **1977**, Ronald Rivest, Adi Shamir and Leonard Adleman developed **RSA**.

RSA became one of the most important public-key algorithms and could be used for:

- Encryption
- Digital signatures
- Key-related operations

Its security is based mainly on the difficulty of factoring large numbers.

4. Earlier Classified GCHQ Work

Earlier classified work at **GCHQ** involved James Ellis, Clifford Cocks and Malcolm Williamson.

- Ellis developed an important concept related to public-key encryption.
- Cocks developed an RSA-like public-key encryption method.
- Williamson developed an approach related to key exchange.

This work remained classified for many years.

5. Later Developments

Other important public-key techniques were developed later, including:

- **ElGamal**
- **Elliptic Curve Cryptography (ECC)**

ECC provides strong security with much smaller keys than RSA.

Q5. *[Advantages and disadvantages]* **[8 Marks]**

Discuss the advantages and disadvantages of asymmetric cryptography. Why are real-world systems such as TLS usually built as hybrid cryptosystems that combine asymmetric and symmetric algorithms? Explain with a suitable example.

Ans- Advantages and Disadvantages of Asymmetric Cryptography

Advantages

1. Easier Key Distribution

The public key can be shared openly. There is no need to secretly send the public key.

2. Digital Signatures

Private keys can be used to create digital signatures for authentication and integrity.

3. Better Scalability

Each user can have one public-private key pair, making large networks easier to manage.

4. Supports Authentication

Public keys can be used to verify digital signatures and identities.

Disadvantages

1. Slower

Asymmetric algorithms are much slower than symmetric algorithms for large amounts of data.

2. Larger Keys

They generally require larger key sizes than symmetric algorithms.

3. More Complex

Key management and certificate systems can be complicated.

Why Hybrid Cryptography Is Used

Real-world systems such as **TLS** combine both types.

Example

1. Asymmetric cryptography is used to authenticate the server and establish or protect a session key.
2. After the secure session is established, a fast symmetric algorithm is used to encrypt the actual data.
3. Large amounts of data can therefore be encrypted efficiently.

Asymmetric cryptography = secure setup
Symmetric cryptography = fast data encryption

Q6. *[RSA algorithms and key generation]* **[12 Marks]**

Describe the RSA algorithm in detail. Explain the key generation process step by step (choosing two large primes, computing the modulus and Euler's totient function, selecting the public exponent, and deriving the private exponent), and then explain how encryption and decryption work, stating the formulas used. Include a flowchart of the algorithm and explain why decryption correctly recovers the original message.

Ans- RSA Algorithm and Key Generation

Introduction

RSA is an asymmetric cryptographic algorithm based on a pair of keys:

- **Public Key:** `(e, n)`
- **Private Key:** `(d, n)`

RSA can be used for encryption and digital signatures.

Step 1: Choose Two Prime Numbers

Choose two large prime numbers:

p and q

For a simple example, small primes can be used for learning, but real RSA uses very large primes.

Step 2: Calculate the Modulus

Calculate:

$n = p \times q$

The value `n` is used in both the public and private keys.

Step 3: Calculate Euler's Totient

Calculate:

$\varphi(n) = (p - 1)(q - 1)$

Step 4: Select Public Exponent

Choose `e` such that:

$1 < e < \varphi(n)$

and

$\gcd(e, \varphi(n)) = 1$

The value `e` becomes part of the public key.

Step 5: Calculate Private Exponent

Find `d` such that:

$d \times e \equiv 1 \bmod \varphi(n)$

The value `d` becomes part of the private key.

Keys

Public Key = (e, n)
Private Key = (d, n)

RSA Encryption

If the message is represented by `M`:

$C = M^e \bmod n$

Here:

- `M` = plaintext message
- `e` = public exponent
- `n` = modulus
- `C` = ciphertext

RSA Decryption

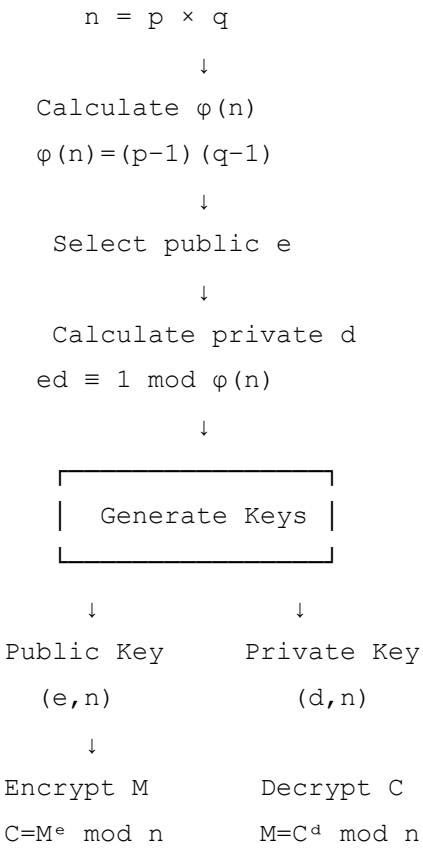
The receiver uses the private key:

$M = C^d \bmod n$

This recovers the original message when RSA is used with the required mathematical conditions.

RSA Flowchart

```
Choose two large primes
        ↓
Calculate n
```



Why Decryption Works

The values `e` and `d` are mathematically related through Euler's totient function. This relationship makes the encryption and decryption operations reverse each other under the RSA conditions.

Q7. *[Key sizes]* **[10 Marks]**

What is meant by the key size of an RSA system? Explain why the security of RSA depends on the difficulty of factoring the modulus. Discuss the commonly recommended RSA key sizes (1024, 2048, 3072 and 4096 bits), compare them with equivalent-strength ECC keys, and comment on the threat that quantum computers (Shor's algorithm) pose to RSA.

Ans- RSA Key Sizes and Security

What is RSA Key Size?

RSA key size refers to the **number of bits in the RSA modulus `n`**.

For example:

- 1024-bit RSA
- 2048-bit RSA
- 3072-bit RSA
- 4096-bit RSA

A larger key generally makes attacks more difficult but also increases computational cost.

Security and Factoring

RSA security is closely connected to the difficulty of factoring:

$n = p \times q$

where p and q are large prime numbers.

An attacker who can efficiently factor n may be able to recover information needed to calculate the private key.

Common RSA Key Sizes

RSA Key Size	General Security View
1024 bits	Too weak for modern long-term security
2048 bits	Common minimum for many current applications
3072 bits	Higher security and longer-term protection
4096 bits	Very large key; stronger margin but slower

RSA and ECC Comparison

ECC provides similar security with much smaller keys.

RSA	Approximate ECC Equivalent
2048-bit RSA	224-bit ECC
3072-bit RSA	256-bit ECC
7680-bit RSA	384-bit ECC
15360-bit RSA	521-bit ECC

The exact security equivalence depends on the security level and algorithm.

Quantum Threat

Large-scale quantum computers could threaten RSA through **Shor's algorithm**. Shor's algorithm can theoretically factor large integers efficiently on a sufficiently powerful quantum computer.

Therefore, RSA is considered vulnerable to future large-scale quantum attacks, which is why **post-quantum cryptography** is an important research area

Q8. *[Security considerations]* **[12 Marks]**

Discuss the major security considerations and known attacks on RSA, including (a) factoring attacks, (b) small public exponent and small message attacks, (c) common modulus attack, (d) timing and other side-channel attacks, and (e) chosen-ciphertext attacks. Explain why padding schemes such as OAEP are required, how RSA is used for digital signatures, and list good practices for secure RSA deployment.

Ans- Security Considerations and Attacks on RSA

RSA is mathematically strong when implemented correctly, but poor key management, incorrect usage and implementation weaknesses can make it vulnerable.

A. Factoring Attacks

RSA security depends on the difficulty of factoring the modulus:

$n = p \times q$

If an attacker can factor n into p and q, they may be able to calculate the private key.

Protection

- Use sufficiently large RSA keys.
- Use strong, randomly generated primes.

B. Small Public Exponent and Small Message Attacks

Using a small public exponent can create problems when RSA is used incorrectly, especially with small messages and improper padding.

Protection

Use secure padding schemes such as **OAEP** instead of directly encrypting raw messages with RSA.

C. Common Modulus Attack

If different users use the **same RSA modulus** with different public exponents, certain conditions can allow an attacker to recover information about the original message.

Protection

Each RSA key pair should have its own properly generated modulus.

D. Timing and Side-Channel Attacks

An attacker may study information leaked by the implementation, such as:

- Execution time
- Power consumption
- Memory behavior
- Other physical or computational patterns

These attacks may reveal information about the private key.

Protection

Use constant-time implementations and trusted cryptographic libraries.

E. Chosen-Ciphertext Attacks

In a chosen-ciphertext attack, an attacker submits specially created ciphertexts and observes how the system responds. Poorly designed RSA encryption can leak information through these responses.

OAEP

RSA-OAEP (Optimal Asymmetric Encryption Padding) adds randomized padding before RSA encryption.

It helps prevent attacks against raw RSA encryption and is the standard approach for RSA encryption in many modern applications.

RSA Digital Signatures

RSA can also be used for digital signatures.

Basic Process

1. The sender creates a hash of the message.
2. The hash is processed with the sender's private key to create a signature.
3. The receiver uses the sender's public key to verify the signature.
4. The receiver also checks the message hash.

This provides:

- **Authentication**
- **Integrity**
- **Non-repudiation**

Good Practices

- Use **2048-bit RSA or stronger** where RSA is required.
- Never use raw RSA encryption for real applications.
- Use **OAEP** for RSA encryption.
- Use appropriate signature padding such as **PSS** for RSA signatures.
- Generate keys using secure random number generators.
- Keep private keys secret and protected.
- Never reuse the same RSA modulus between unrelated key pairs.
- Use trusted, well-tested cryptographic libraries.
- Protect private-key operations against side-channel attacks.
- Plan for migration to **post-quantum cryptography** for long-term security.